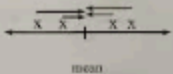


Definitions	Examples
Lesson 8.1.3 Mean Absolute Deviation	<u>Mean Absolute Deviation</u> Average distance each point is from the mean.  mean
Lesson 8.1.3 Learning Log Mean Absolute Deviation Date:	
Lesson 8.1.4 Quartiles and Interquartile Range (IQR)	<u>Quartiles</u> Points that divide data into four equal parts 2 7 9 13 lower half ↓ 2 7 9 13 ↓ 8 first quartile (median of the lower half) 14 ↓ median 15 22 26 43 upper half ↓ 22 26 36 43 ↓ 34 third quartile (median of the upper half) <u>Interquartile Range (IQR)</u> Range from 1st Quartile to 3rd Quartile

Record any definitions, examples, or formulas for each section

Definitions	Examples
Lesson 8.1.2 Measures of Central Tendency	Mean: Average Outliers: Numbers larger or smaller than the main others Median: Middle number when arranged numerically Mode: Number that shows up the most Example: 10, 10, 10, 10, 10, 8 9, 9, 13, 15, 16, 21, 26 Mean: 15.57 Median: 15 Mode: 9
Lesson 8.1.2 Learning Log Mean and Median Date: _____	

$$\begin{array}{r}
 36 \\
 26 \\
 16 \\
 15 \\
 13 \\
 9 \\
 \hline
 109
 \end{array}
 \begin{array}{r}
 15.57 \\
 7 \overline{) 109.00} \\
 \underline{-70} \\
 39 \\
 \underline{-35} \\
 40 \\
 \underline{-35} \\
 50 \\
 \underline{-45} \\
 5
 \end{array}$$

8-12. Problem continued from previous page.



5. What was more obvious when you looked at the histograms compared to looking at the list of data? Is there information that is easier to see on the histograms?
[Some students may notice that the 2008 data is more balanced or symmetrical, while the 2009 data has outliers.]
6. The technology tool shows the mean and the median of the data below the histograms. However, if the mean and the median were not labeled, would there be a way to determine the median or mean from a histogram alone? Explain.
[No, it is not possible, since the histogram only gives frequency of data in a range of each bin, and does not provide information about the individual data points.]

8-13. Use the list of data in problem 8-12 or the histograms that you or your teacher made to continue to analyze and compare the frog jumps for 2008 and 2009. Answer the following questions.

- a. What was the range of the jumps in each year? What does this tell you about the frog jumps? [The data for 2008 fall between 216.5 inches and 231.5 inches, so the range is 15 inches. The data for 2009 fall between 185.25 inches and 252 inches, so the range is 66.75 inches. It shows that the frogs jumped more consistently in 2008.]
- b. What was the typical jump length of the frogs each year? How did you find this value? [See the "Suggested Lesson Activity"; 2008 mean = 224.125, 2008 median = 224.25, 2009 mean = 225.1, 2009 median = 225.]
- c. Were the jumps all about the same, or were some jumps outliers? Name any outliers and explain why you think they are outliers. [The 2009 data has two outliers: 185.25 and 252 inches.]
- d. Compare as completely as you can the 2008 jumps to the 2009 jumps. Compare the center, shape, spread (range), and outliers. Then draw a conclusion: were one year's frogs a better group of jumpers than the other? How do you know?
[Sample response: As a group, the frogs in 2009 jumped farther. However, the frogs in 2008 jumped more consistently.]

2008 mean jump	224.125
2009 mean jump	225.1
2008 median jump	224.25
2009 median jump	225

8.1.2 What is a typical value?

Choosing Mean or Median

You are exposed to a huge amount of information every day in school, in the news, in advertising, and in other places. It helps to have tools to be able to understand what the data means. Today you will turn your attention to what you can learn from data.

8-12. JUMPING FROG JUBILEE

In Lesson 1.1.4, you examined results of the Calaveras County frog-jumping contest. You saw that the 2009 competition winner jumped 20.5 inches farther than the 2008 winner. Do you think that the winner in 2009 was a special frog? Or were the frogs in 2009 better jumpers, in general, than the frogs in 2008?



With your team, brainstorm ways that you could compare the two groups of frogs. Recall some of the measures of central tendency from the previous lesson.

2008		2009	
Frog Name	Jump Length	Frog Name	Jump Length
Skinner Eater	231.5 in.	For the Sign	252 in.
Warped	210 in.	Alex Frog	216.5 in.
Grog Croone Deme	229 in.	Shakin	231.5 in.
R.D.	227 in.	Six Mile Shooter	226.75 in.
The Well Ate 1 Day	221.5 in.	Squash the Air Every Day	223.25 in.
Winner	220.5 in.	Huapet	223.25 in.
7 lb 8 oz. Baby	217 in.	Jennifer's Jumper	222.25 in.
Delbert Sr.	216.5 in.	Dr. Frog	185.25 in.

- a. Your first job is to make a graphical representation of the data. Many statisticians say that the first and most important step in analyzing any data is to make a graphical representation. How can a representation help you analyze the data? Use the *Representing Data* technology tool (available at www.cpm.org/students/technology) to create histograms of the data, or look at the ones that your teacher displays for you. The bar size of each histogram should be 10 inches. Why is a histogram a good choice? [A graphical representation allows us to see more connections between and patterns in the data, such as spread and shape. See the following example histograms. A histogram, with bin widths of 10 in., is a good choice here because the data is numerical and values are not repeated. In addition, a dot plot would stretch from 185 to 252 with increments of 0.01. There would only be one data value at each of 8 points along this very long dot plot.]



Problem continues on next page. →
Core Connections, Course 1